

C1M4_UngradedLab_1

September 30, 2026

1 Module 4 Ungraded Lab 1: Improper Prior

Ungraded labs are designed to give learners an opportunity to apply theoretical material presented in previous videos. The goal of this ungraded lab is to utilize an improper prior to conduct Bayesian inference in R. Ungraded labs can be completed when assigned or in conjunction with upcoming videos. Answers to ungraded labs will be provided in upcoming videos.

1.1 Problem #3 Improper Priors

Let $X_1, \dots, X_n \stackrel{iid}{\sim} N(\mu, 1)$ and let $\pi(\mu) = 1/10$ for $\mu \in (-\infty, \infty)$. Clearly, $\int_{-\infty}^{\infty} \pi(\mu) d\mu > 1$, so $\pi(\mu)$ is improper.

(a) Show numerically that the posterior is proper.

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(b) Plot the prior, likelihood, and posterior on the same plot.

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(c) Provide numerical evidence that the maximizer of the posterior is equivalent to the MLE.

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